

Homework 4- Due Date Wednesday April 2, 11:59 P.M.

Problem 1. Find the volume under the surface $z = \frac{\ln(x) \ln(y)}{4xy}$ and above the rectangular region $1 \leq x \leq e^2, 1 \leq y \leq e^2$.

Problem 2. Suppose a discrete random variable K has the following pmf:

$$p_K(k) = \frac{1}{2^k} \quad ; k = 1, 2, 3 \dots$$

- (a) Find cumulative distribution function $F_K(k)$.
- (b) Find $E[2K - 1]$.

Problem 3. Assume pdf of a random variable X is given as below:

$$f(x) = \begin{cases} \frac{1}{(b-a) + (d-c)} & x \in [a, b] \cup [c, d] \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

- (a) Find cumulative distribution function $F_X(x)$.
- (b) Find $\text{Var}(X)$.

Problem 4. There are two coins, one with probability p_1 of Heads and the other with probability p_2 of Heads. One of the coins is randomly chosen (with equal probabilities for the two coins). It is then flipped $n \geq 2$ times. Let X be the number of times it lands Heads.

- (1) What is the PMF of X ? (Hint: Find what values X can take and for each $X = x$ find $P(X = x)$)
- (2) What is distribution of X when $p_1 = p_2$.

Problem 5. Sarah goes for shopping cookies to a bakery that sells n different types of cookies. Sarah samples a cookie with probability p (independently for each type). If a cookie type is sampled, then she buys that type with

probability q , independently of other cookie types. Let Y be the number of cookie types that are sampled, and let X be the number of cookie types that Sarah buys. Find the distribution of Y and the distribution of X . Explain your answer.

Problem 6. Sarah immigrated to a new city, and would like to make friends with her new neighbours.

On any particular day i , she feels shy with probability $1 - p$ ($0 < p < 1$) and stays home; or, with probability p , she goes out and visits the i -th house in her neighborhood. At any house that she visits, either:

- (i) someone is at the house and answers the door; this happens with probability q (where $0 < q < 1$). In that case, Sarah shows them a picture of her hometown;
- (ii) no one is at the house and Sarah returns home.

We assume that the collection of all events of the form Sarah stays home on day i and Someone is at the i -th house on day i , for $i = 1, 2, \dots$, are (mutually) independent.

- (6-1) What is the probability that Sarah gives out a picture on the first day?
- (6-2) What is the probability that she gave out exactly k pictures on the first n days? where k and n are some fixed integers with $n \geq 1$ and $0 \leq k \leq n$.
- (6-3) What is the probability that she stayed home on day 1, given that she did not give out a picture on that day?
- (6-4) Within the first week (7 days), what is the probability that there were no days in which Sarah went out but did not give out a picture of her hometown?